

SENG ATHIT BANNAVONG

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EDUCATION

Hanoi University of Science and Technology

Bachelor of Computer Science

GPA: 2.33 / 4.00

2023 – Present

Expected Graduation: 2027

Phiawat High School

Graduated with Distinction

2019 – 2022

Grade: 40.05 / 50

LANGUAGES

Lao: Native **English:** B1 (Intermediate) **Vietnamese:** B1 (Intermediate) **Thai:** B1 (Intermediate)

SKILLS

Programming: C/C++, Java, JavaScript, Python, Rust (learning)

Web: React, Node.js, MySQL

Tools & Platforms: Git, Docker, WSL2, Linux, VS Code, JetBrains, Figma

PROJECTS

Text Classification using TF-IDF and Naive Bayes

University Project, 2025

github.com/SengathitBannavong/NLP_TFIDF_NaiveBayes_From_C

- Built a complete NLP text classifier **from scratch in C** with no external ML libraries
- Implemented TF-IDF vectorizer, Naive Bayes classifier, sparse matrix, string pool, and word hashing manually
- Used multithreaded preprocessing to handle large text datasets efficiently (48 commits)

Thread Pool + Priority Task Scheduler

Own Project, 2026

github.com/SengathitBannavong/thread_pooling

- Implemented a thread-safe priority task queue and worker pool in C using POSIX threads
- Built graceful shutdown, blocking/non-blocking pop, and mutex + condition variable synchronization
- Integrated ThreadSanitizer (TSAN) into CI pipeline; all concurrency tests pass cleanly

Transcript Viewer

Own Project, 2026

github.com/SengathitBannavong/Transcript-Viewer

- Desktop transcript viewer in C (Clay UI + Raylib + SQLite3) with donut chart, CPA gauge, graduation checklist, and command palette (Ctrl+K)
- 49 unit tests covering GPA, credit, and graduation logic; cross-platform build (Linux + Windows)

Vietnamese Text Recognition System

University Project, 2025

github.com/SengathitBannavong/Recognized-Vietnam-Sentence

- Built an end-to-end OCR pipeline for Vietnamese text using CRNN (CNN + BiLSTM + CTC) trained on synthetic data
- Implemented full pipeline: synthetic dataset generation (TRDG), LMDB preprocessing, model training on Colab GPU, and Flask web app for inference

Web E-Commerce Platform

University Project, 2025

github.com/SengathitBannavong/Web-E-commerce

- Full-stack book store with React + Tailwind frontend, Express + PostgreSQL + Redis backend, Docker, and CI/CD pipeline; team of 3

EXPERIENCE

Video Editing Assistant

2023

- Assisted a teacher in editing educational videos, vlogs, and lecture recordings

Teaching Assistant (Computer Class)

Phiawat High School, 2019

- Taught basic computer skills (Word, Excel, PowerPoint) to secondary school students
- Assisted with classroom computer setup and software maintenance

CERTIFICATIONS & AWARDS

- Introduction to MS Office 2007 — Lao Child Development Center (2013)
- Scholarship Award — Logo Academy (2019)